

Quantum dynamics in real Hilbert space: Algebraic isomorphism and symplectic geometry of the Schrödinger equation

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Highlights

- Synthesizes a rigorous real Hilbert space framework for quantum dynamics.
- Constructs explicit real representations for multi-qubit state vectors.
- Derives a fully real-valued protocol for generating entangled Bell states.
- Maps unitary quantum gates to real symplectic and orthogonal matrices.
- Refines the quantum axiomatic framework using compatible symplectic structures.

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Abstract

This work reformulates quantum mechanics in real Hilbert space and demonstrates that the complex structure inherent to the Schrödinger equation can be fully captured by a coupled system of real-valued dynamics. By employing the standard field isomorphism of $(\mathbb{R}^2, +, \cdot)$ to $\mathbb{C}$, the real counterpart of the Schrödinger equation is derived, yielding a system of coupled real partial differential equations. This system reveals a symplectic structure where the symplectic matrix $\mathcal{J}$ geometrically represents the imaginary unit i , acting as a phase-space rotation operator. Time-independent solutions decouple into identical real Helmholtz equations, preserving spectral equivalence. The reformulation provides geometric insights into foundational aspects: the geometric representation of i , the emergence of probability conservation from real-component interactions, and the encoding of quantum phase relationships through the multiplicative structure of the $\mathbb{R}^2$ field. This perspective reinforces connections to classical Hamiltonian mechanics and geometric quantization, suggesting that complex numbers are structural, emerging from field operations on the real plane, rather than fundamental. Building on these foundations,

we present original results where quantum computation elements like qubits and gates are shown to have equivalent representations as real symplectic and special orthogonal matrices, maintaining unitary equivalence. We introduce a framework for quantum computation within a real vector space and provide a generalized protocol for generating entangled Bell states. These results offer a new perspective on the necessity of complex numbers in quantum theory, providing a framework with potential computational advantages for hybrid quantum–classical systems and enhanced geometric interpretability.

Keywords

Field isomorphism; Quantum mechanics; Schrödinger equation; Unitary-real symplectic and special orthogonal transformations; Quantum gates and quantum computing; Superposition; Entanglement and Bell states

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Introduction

The complex numbers $\mathbb{C}$ hold a central and historically significant role in the mathematical formulation of quantum mechanics, notably through their appearance in the Schrödinger equation [1]. This ubiquitous presence, despite its computational utility, has spurred enduring foundational inquiries into the ontological necessity of complex numbers versus their interpretation as a representational convenience [2].

This study addresses this fundamental question within the algebraic framework by leveraging the field isomorphism that exists between the complex numbers $\mathbb{C}$ and the real plane $\mathbb{R}^2$, provided the latter is endowed with suitably defined algebraic operations. The field $(F, +, \cdot)$ is characterized by the properties that $(F, +)$ forms an abelian group, $(F \setminus \{0\}, \cdot)$ constitutes an abelian group, and multiplication distributes over addition [3], [4]. We utilize the standard algebraic result that $\mathbb{R}^2$ can be endowed with a field structure isomorphic to $\mathbb{C}$ through the following definitions [5], [6], [7]:

$$(a, b) + (c, d) := (a + c, b + d), \quad (a, b) \cdot (c, d) := (ac - bd, ad + bc).$$

The isomorphism is formally established by the mapping $\mu : \mathbb{C} \rightarrow \mathbb{R}^2$, defined as $\mu(a + ib) = (a, b)$, which preserves the defined field operations. This algebraic equivalence naturally leads to a real-valued formulation of quantum dynamics, wherein quantum states are represented by $\mathbb{R}^2$ -valued wavefunctions $\Psi(\mathbf{r}, t) = (u(\mathbf{r}, t), v(\mathbf{r}, t))$, and real-valued physical parameters are embedded into $\mathbb{R}^2$ through the injective mapping $\iota : \mathbb{R} \hookrightarrow \mathbb{R}^2$, given by $\iota(a) = (a, 0)$.

Under the isomorphism μ , the time-dependent Schrödinger equation [8] is transformed into a coupled system of partial differential equations:

$$-\hbar \partial_t v = -\frac{\hbar^2}{2m} \nabla^2 u + V u, \quad \hbar \partial_t u = -\frac{\hbar^2}{2m} \nabla^2 v + V v.$$

This transformation reveals an underlying symplectic structure, expressed as [9]:

$\hbar \partial_t \mathbf{U} = \mathcal{J} \left(-\frac{\hbar^2}{2m} \nabla^2 + V \right) \mathbf{U}$, where $\mathbf{U} = \begin{pmatrix} u \\ v \end{pmatrix}$ and $\mathcal{J} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$. This formulation

provides a geometric interpretation of quantum phase space, consistent with contemporary paradigms in geometric quantization [10]. Our analysis further demonstrates that: (i) The imaginary unit i is operationally equivalent to the symplectic rotation induced by the field product $(0, 1) \cdot (u, v) = (-v, u)$, thereby elucidating its role in phase evolution [11]. (ii) The principle of probability conservation, expressed as $\partial_t (u^2 + v^2) + \nabla \cdot \mathbf{j} = 0$, arises directly from the dynamics of the real-valued components [12]. (iii) Spectral equivalence is maintained: the time-independent solutions of the real-valued Schrödinger equation decouple into two identical real Helmholtz equations, ensuring the preservation of eigenvalue degeneracy [13], [14].

While demonstrably equivalent to the standard complex formulation in terms of physical predictions, this real-valued perspective sheds light on the underlying geometric foundations of quantum theory. The doubled real degrees of freedom (u, v) inherently encode phase relationships through the defined field product (1), suggesting that the role of complex numbers extends beyond mere computational convenience to a structural necessity within the standard formalism. This work establishes a connection between algebraic and geometric approaches to quantization, potentially offering novel perspectives on the long-standing problem of quantum–classical correspondence [8].

The decomposition of the Schrödinger equation into real and imaginary components is a well-established concept, with origins tracing back to the early development of quantum mechanics in 1926 [1]. This approach has been explored extensively in the literature to investigate the geometric and symplectic structures underlying quantum dynamics. A significant geometric framework, advanced by Volovich, recasts the Hilbert space of quantum states as a real Kähler manifold [15], [16]. This formulation, endowed with compatible symplectic and Riemannian structures, provides a unified setting for quantum unitary evolution and classical Hamiltonian dynamics. By rigorously analyzing the structural relationship between complex and real quantum mechanics, this approach demonstrates how quantum operations and entanglement can be equivalently expressed in real symplectic terms, thereby consolidating the mathematical foundations of real-valued quantum theory.

The material developed in Sections 3 Schrödinger equation in the complex field, 4 Schrödinger equation in the, 5 Structural analysis, 6 Equivalence between, 7 Comparative analysis, 8 Quantum mechanics in a real symplectic-special orthogonal framework synthesizes these foundational concepts into a unified and self-consistent mathematical framework to support the subsequent developments. While the core dynamical principles in these sections review established theory, this work offers the following specific contributions: (i) a constructive unification of distinct realification conventions, including the explicit derivation of a permutation matrix that connects them; (ii) a comprehensive correspondence table that systematically aligns the algebraic, dynamical, spectral, and symmetry structures of the conventional complex formulation with those of the real symplectic–orthogonal representation; (iii) the refinement of the axiomatic framework for quantum theory defined on a real vector space endowed simultaneously with symplectic and orthogonal structure; and (iv) the substantive original extension of this framework to quantum information, providing explicit real-only constructions of qubit gates, circuit-composition rules,

and a fully real protocol for generating Bell states. These developments collectively advance the real-space formulation from a theoretical curiosity to a practical computational framework.

The role of complex numbers within the mathematical framework of quantum mechanics has been a long-standing subject of profound foundational debate. The central question revolves around whether their presence signifies an inherent ontological necessity or merely represents a convenient mathematical tool for calculations. Adherents to the indispensability of complex numbers, as highlighted by recent investigations [17], [18], [19], [20], contend that the complex structure is an intrinsic feature of quantum theory. They point to its fundamental appearance in the Schrödinger equation, the principles of superposition and interference phenomena, and the description of experiments conducted on elementary quantum networks as compelling evidence. This perspective posits that complex numbers are essential for encoding fundamental phase relationships and non-classical correlations that cannot be adequately captured by formalisms based solely on real numbers without compromising physical consistency. Conversely, this view has been challenged by researchers, including the early works of Birkhoff and von Neumann [21], and Einstein [22], who suggest that complex numbers might be a representational artifact linked to the Hilbert space formulation of quantum mechanics. Furthermore, advancements in geometric quantization [23], [24], [25], [26], [27], [28], [29] have demonstrated the possibility of describing quantum dynamics using real symplectic structures, thereby questioning the notion of an ontologically privileged status for complex numbers. The current work addresses this debate by explicitly employing the standard algebraic isomorphism between the field of complex numbers $\mathbb{C}$ and the real plane $\mathbb{R}^2$. This construction demonstrates that complex arithmetic arises structurally from the field operations defined on $\mathbb{R}^2$. The proposed reformulation preserves all physical predictions of standard quantum mechanics — including probability conservation and spectral equivalence — while interpreting the imaginary unit i as a symplectic operator $\mathcal{J}$, effectively disentangling quantum mechanics from an *a priori* reliance on complex numbers. These findings suggest that while computationally advantageous, complex numbers are not fundamentally ontological but rather emerge from the geometric structure of $\mathbb{R}^2$, paving the way for a paradigm where quantum reality is potentially grounded in real symplectic dynamics.

It is important to address historical arguments concerning the viability of formulating quantum mechanics on a real Hilbert space. Seminal work by Stueckelberg [26] and more recent geometric analyses by Moretti and Oppio [29] have shown that standard quantum mechanics cannot be consistently formulated on a real Hilbert space without introducing an additional operator structure J , satisfying $J^2 = -I$, which commutes with the full algebra of physical observables. Such a structure effectively reinstates the complex structure at the algebraic level. The present work does not contradict these results; rather, it makes this structure explicit by identifying J with the canonical symplectic matrix $\mathcal{J}$. By interpreting the symplectic structure as the geometric realization of the imaginary unit, the framework satisfies the algebraic requirements identified in [26], [29] while maintaining a strictly real-valued coordinate representation.

The Bell states, a set of four maximally entangled two-qubit states, are foundational to the study of quantum entanglement, representing a quintessential example of non-classical correlations that classical physics cannot explain [30]. These states exhibit perfect correlations or anti-correlations between the measurement outcomes

of distant parties; however, only the singlet state exhibits perfect anticorrelations along any arbitrary direction, while for other Bell states, these correlations depend on the specific measurement axis. The significance of Bell states is intrinsically linked to the Bell inequality, a mathematical constraint derived under the assumptions of local realism and causality. Local realism posits that physical properties have definite, pre-existing values that are independent of measurement, and that physical influences are limited by the speed of light due to the principle of locality. The Bell inequality provides an upper bound on the correlations that can be observed in any physical theory satisfying these principles. However, quantum mechanics predicts correlations for certain entangled states, including Bell states, that violate this bound. Numerous experiments have conclusively demonstrated these violations, providing compelling evidence against local hidden variable theories [31] and strongly supporting the violation of local realism by certain entangled systems as predicted by quantum mechanics. This study introduces a framework for quantum computation within a real vector space and presents a generalized protocol for generating entangled Bell states.

The main contributions of this work are as follows:

1. **Mathematical Foundations:** We synthesize the realification functor, embedding of the unitary group into the real symplectic group, unitary-real symplectic and special orthogonal equivalence, Hermitian-real symmetric equivalence, Hermitian-unitary matrices characterization, expectation values equivalence under realification, and fundamental axioms of quantum mechanics within the real symplectic and special orthogonal framework.
2. **Algebraic Isomorphism Between $\mathbb{C}$ and $\mathbb{R}^2$:** We employ the standard field isomorphism between the complex numbers $\mathbb{C}$ and the real plane $\mathbb{R}^2$ equipped with appropriately defined operations, demonstrating the structural emergence of complex arithmetic from real-valued operations.
3. **Real Reformulation of Quantum Dynamics:** We formalize the symplectically coupled system of real partial differential equations equivalent to the Schrödinger equation, demonstrating a complete description of quantum evolution without explicit reliance on complex numbers while preserving dynamical equivalence.
4. **Symplectic Geometric Foundation:** We characterize the symplectic matrix $\mathcal{J}$ as the geometric realization of the imaginary unit i , unifying Hamiltonian mechanics with quantum phase-space dynamics and elucidating the role of symplectomorphisms in maintaining probabilistic structure.
5. **Spectral and Conservation Equivalence:** We verify that time-independent solutions decompose into identical real Helmholtz equations, preserving spectral degeneracy, and derive probability conservation $\partial_t (u^2 + v^2) + \nabla \cdot \mathbf{j} = 0$ directly from the dynamics of the real-valued components.
6. **Quantum Computation in Real Space:** We introduce a novel framework for quantum computation within a real vector space, *demonstrating a*

generalized protocol for generating entangled Bell states. This involves a systematic mapping of n -qubit states from the complex Hilbert space $\mathbb{C}^{2^n}$ to vectors in $\mathbb{R}^{2^{n+1}}$, and of unitary quantum gates from $\mathbb{C}^{2^n \times 2^n}$ to real symplectic and special orthogonal matrices within $\mathbb{R}^{2^{n+1} \times 2^{n+1}}$. This mapping establishes a complete equivalence with standard complex-valued quantum computing formalisms, thereby enabling simulations and computations based entirely on real numbers.

7. **Resolution of Foundational Debates:** We present a perspective suggesting that complex numbers $\mathbb{C}$ in quantum theory are not ontological necessities but rather structural consequences of $\mathbb{R}^2$ field operations, potentially resolving long-standing philosophical debates and bridging geometric quantization with classical mechanics.
8. **Computational and Interpretational Advancements:** We propose a framework for hybrid quantum–classical models, topological analysis via real-space geometry, and algorithmic simplification by explicitly separating amplitude and phase degrees of freedom.

These contributions collectively redefine the mathematical foundations of quantum mechanics, offering geometric clarity, computational practicality, and deeper insight into the interplay between algebraic structures and physical reality.

Let μ denote the canonical realification operation, with instances: (i) $\mu : \mathbb{C} \rightarrow \mathbb{R}^2$ (scalars), (ii) $\mu : \mathbb{C}^n \rightarrow \mathbb{R}^{2n}$ (vectors), (iii) $\mu : M_n(\mathbb{C}) \rightarrow M_{2n}(\mathbb{R})$ (matrices), (iv) $\mu : \mathcal{H} \rightarrow \mathcal{S}$ (Hilbert to symplectic). All instances preserve algebraic, dynamical, and observable structures.

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Section snippets

Mathematical foundations

First, we rigorously construct a field structure on the vector space $\mathbb{R}^2$ by defining appropriate addition and multiplication operations. This structure is foundational for understanding complex numbers algebraically. ...

Field-theoretic formulation

Let $\mathbb{C}$ be the field of complex numbers with its standard operations. We consider quantum states represented by $\mathbb{C}$ -valued wave functions defined on spacetime:

Definition 6 Wave Function

A *wave function* is a complex-valued function $\Psi : \mathbb{R}^3 \times \mathbb{R} \rightarrow \mathbb{C}$ such that, for each fixed time $t \in \mathbb{R}$, the spatial slice $\Psi(\cdot, t)$ belongs to the Hilbert space $L^2(\mathbb{R}^3; \mathbb{C})$. For the differential formulation of the Schrödinger equation, we further require $\Psi(\cdot, t) \in H^2(\mathbb{R}^3)$, the standard Sobolev space. Physical constants are canonically embedded into $\mathbb{C}$ via the ...

...

Field-theoretic formulation

Let the quantum state be represented by an $\mathbb{R}^2$ -valued wave function

$\Psi : \mathbb{R}^3 \times \mathbb{R} \rightarrow \mathbb{R}^2$, $\Psi(\mathbf{r}, t) = (u(\mathbf{r}, t), v(\mathbf{r}, t))$. For each fixed time t , the components satisfy $u(\cdot, t), v(\cdot, t) \in H^2(\mathbb{R}^3)$, the Sobolev space of square-integrable functions with square-integrable first and second spatial derivatives, ensuring that the spatial Laplacian in the Schrödinger operator is well defined. In addition, the time evolution is governed by the Schrödinger equation, which guarantees the existence of the first time derivatives $\partial_t u$...

Symplectic structure

The time-dependent system (33) can be expressed in Hamiltonian form. Letting $\mathbf{U} = (u, v)^\top$, we have:

$$\hbar \partial_t \mathbf{U} = \mathcal{J} \left(-\frac{\hbar^2}{2m} \nabla^2 + V \right) \mathbf{U},$$

where

$$\mathcal{J} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad \mathcal{J}^2 = -\mathbb{I}, \quad \mathcal{J}^\top = -\mathcal{J}$$

is the symplectic matrix. This reveals the inherent symplectic geometry underpinning the $\mathbb{R}^2$ formulation.

The time-dependent system (33) admits a natural Hamiltonian formulation when viewed through the lens of symplectic geometry. Recall that a symplectic manifold is a pair (M, ω) , where M is a smooth manifold and ω is a closed, non-degenerate 2-form [9]. In our ...

Equivalence between $\mathbb{C}$ and $\mathbb{R}^2$ formulations

This section outlines the structural correspondence between the complex ($\mathbb{C}$) and real ($\mathbb{R}^2$) formulations of quantum mechanics, detailing the algebraic foundations, state spaces, dynamical principles, and symmetry groups in each framework, along with the mechanisms establishing their equivalence. The fundamental equivalence

between the complex and real formulations of quantum mechanics is rigorously established through the following categorical comparison:

Proposition 2

The formulations are physically equivalent ...

...

Comparative analysis

This section presents a detailed comparative analysis between complex and real formulations as follows. ...

Quantum mechanics in a real symplectic-special orthogonal framework

This section presents a rigorous reformulation of standard complex quantum mechanics by embedding the complex Hilbert space structure into a real vector space endowed with both symplectic and orthogonal structures. This framework establishes a direct, structure-preserving correspondence between unitary operators on complex Hilbert spaces and a specific subgroup of real linear transformations characterized by the preservation of both structures. ...

Qubit representation and quantum gates in $\mathbb{R}^2$

We introduce a representation of qubits and quantum gates in the real vector space $\mathbb{R}^2$ that is isomorphic to the conventional complex number formulation in $\mathbb{C}$. This approach allows us to express quantum states and operations using real-valued vectors and matrices. This work details this $\mathbb{R}^2$ representation and demonstrates its equivalence to standard quantum gates.

A qubit quantum gates in complex $\mathbb{C}$ representation are unitary operators $U \in \text{U}(2)$, and quantum states $|\psi\rangle$ are complex unit vectors. In the ...

Construction of entangled Bell states in the $\mathbb{R}^2$ formulation

We present a systematic method to generate the four maximally entangled Bell states [30] within the real symplectic framework of $\mathbb{R}^2$ quantum mechanics. This approach leverages the isomorphism between $\mathbb{C}$ and $\mathbb{R}^2$ to express entanglement using real-valued vectors and symplectic transformations. ...

Generalization to n -qubit systems

We have established the real representation for a single qubit in $\mathbb{R}^4$ (equivalent to $\mathbb{C}^2$) and illustrated the two-qubit case in $\mathbb{R}^8$ (equivalent to $\mathbb{C}^4$) using the entrywise realification map $\tilde{\mu}$, we now generalize this framework to systems comprising an arbitrary number of n qubits. ...

Mathematical predictions for extended frameworks

The real space formulation of quantum mechanics, as presented in this work, is rigorously demonstrated to be mathematically and physically equivalent to the standard complex Hilbert space formulation. Consequently, the primary

mathematical predictions derived within this real framework precisely coincide with those of conventional quantum mechanics. These include, but are not limited to:

1. The probabilities of outcomes for any given measurement on a quantum system. As shown, the probability $P(\lambda_i)$...

...

Conclusion

This work has systematically reformulated quantum mechanics in the real Hilbert space, demonstrating that the complex structure of the theory arises intrinsically from algebraic and geometric properties of a tailored field structure. By constructing an isomorphism $\mu : \mathbb{C} \rightarrow \mathbb{R}^2$ and endowing $\mathbb{R}^2$ with component-wise operations, we have shown that the Schrödinger equation and its physical predictions are fully recoverable through a system of coupled real-valued partial differential equations. The ...

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